

Transferrable contextual feature clusters for parking occupancy prediction

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ABSTRACT

Recently, real-time parking availability prediction has attracted much attention since the rapid development of sensor technologies and urbanisation. Most existing works have applied various models to predict long and short-term parking occupancy using historical records. However, historical records are not available for many real-world scenarios, such as *new urban areas*, where parking lots are fast adjusted and extended. In this paper, we aim to predict parking occupancy using historical data in other areas and contextual information within the targeted area that lacks historical data. We propose a two-step framework to first learn the important contextual features from areas where parking records already existed. Then we transfer these features to the other new areas without historical data records. Through conducting a real-world dataset with various clustering methods combined with different regression models, we observe that multiple contextual features are likely to influence parking availability prediction. We find the best combination (i.e., *k*-shape clustering algorithm and LSTM regression model) to build parking occupancy prediction model based on the subsequent quantitative correlation analysis between contextual features and parking occupancy. The experimental results show that (1) the conventional internal clustering evaluation does not work well for spatio-temporal data clustering for the prediction purpose; (2) our proposed approach achieves approximately 3% error rate in 30 minutes of prediction, which is significantly better than the estimation of the occupancy rate using the rate in the adjacent regions (13.3%).

1. Introduction

Smart parking is a crucial element in the development of smart cities. The act of searching for available on-street parking can result in negative outcomes [1] such as economic losses, increased air pollution, and added stress for drivers. Research [2] has shown that drivers spend an average of 8.1 min cruising for parking spots, which leads to wastage of time and fuel [3], higher fuel consumption, and air pollution. Additionally, cars cruising for parking contributes to traffic congestion and can slow down the average speed of other vehicles on the road [4]. Therefore, reducing the time spent cruising for parking is vital for smart city management and environmental protection.

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Parking availability prediction is an essential component of smart parking services that provides real-time parking occupancy information. This prediction can significantly mitigate the negative effects of parking cruising and improve the user experience for drivers by reducing the time spent searching for parking spaces and reducing stress levels. Moreover, it provides valuable insights to parking administrators, enabling them to optimise parking management strategies, such as adjusting parking fees based on the estimated parking occupancy to improve the overall efficiency of on-street parking resources [5]. Parking availability prediction based on usage context features can facilitate the expansion and adjustment of parking lots in new urban areas. This can help ensure that parking resources are available and accessible to drivers, reducing the time and stress associated with finding parking. Additionally, it can help reduce the negative environmental and economic impacts of cruising for parking by optimising parking resources to meet demand.

Previous studies on parking availability prediction have mostly used statistical or data-driven approaches based on historical data records to generate short-term parking occupancy predictions. Many of these studies have assumed that parking events follow a Poisson [6] or negative exponential distribution [7,8] based on the arrival and departure processes of drivers. However, these models may not be accurate in the real world due to various contextual features that influence drivers' parking behaviours. Data-driven models have become popular due to the development of machine learning and deep learning techniques [9–11], as well as the availability of in-ground sensor data [12,13]. However, they still heavily rely on historical data, limiting their utility in scenarios where such data is scarce or nonexistent.

To address this issue, Yang et al. [14] presented a deep learning-based framework that incorporates multiple contextual information sources for predicting parking occupancy. However, predicting parking occupancy in a new urban area using only contextual information remains an open research problem. Jayarajah et al. [15] proposed a method to predict the usage patterns of parking lots using contextual features, but this method lacked research on clustering methods and an analysis of features. Furthermore, the approach focused on predicting usage patterns rather than future occupancy rates, making it less practical.

The task of predicting parking occupancy for a new urban area is challenging due to various reasons. Firstly, data-driven methods face difficulty in predicting parking occupancy as historical data is not available for the new area. Secondly, existing works tend to predict parking occupancy for all blocks or grids in a map, which is not practical for administrators or drivers who are more interested in specific areas near their destination. Hence, parking areas need to be divided into sub-areas to explore the influence of contextual information in each sub-area. However, grouping parking slots and evaluating performance without external labels remain unsolved problems. Thirdly, selecting contextual features is challenging as the influence range of each point of interest (POI) and its features are difficult to measure. Therefore, there is a need for a novel solution to predict parking occupancy using only contextual information in new urban areas.

In this paper, we propose a solution to predict parking occupancy in a new urban area without historical data by conducting an empirical study on spatio-temporal contextual features using a real-world dataset from the Melbourne open data platform [16].

Our proposed solution consists of a two-step framework. In the first step, contextual features are extracted by comparing parking occupancy in different sub-areas. This is achieved by grouping parking slots with similar trends using a shape-based clustering method. It is assumed that parking occupancy for each parking lot has similar trends if contextual features are similar [9]. In the second step, the occupancy of each cluster is predicted using regression models and historical occupancy data from each cluster. This approach allows for the prediction of parking occupancy in a new area without relying on historical data, by utilising contextual information from similar sub-areas in the same city. The evaluation of clustering results is based on prediction accuracy since the purpose of clustering is to estimate parking occupancy more accurately. We also conduct a statistical study on various contextual features and analyse their correlation with parking occupancy in Melbourne. Our proposed solution can be generalised to other big cities, as most features are also present in other cities and generally influence parking behaviours.

We also found that traditional internal clustering evaluation metrics do not work well for spatio-temporal data clustering when predicting parking occupancy. Therefore, we evaluated the clustering results using both internal evaluation metrics and prediction accuracy. Our findings showed that the shape-based clustering method with regressors performed better than other clustering methods, such as admin region division, in terms of prediction accuracy. However, the classic internal clustering evaluation metrics showed the opposite result. This suggests that traditional internal clustering evaluation metrics may not be suitable for evaluating spatio-temporal data clustering when prediction accuracy is the primary goal.

Our contributions are as follows:

- We proposed a two-step framework to predict parking occupancy using only contextual features without historical data. We proposed a transferring method to learn the importance of contextual features from the area with historical data and apply them to the new areas.
- We performed an in-depth analysis of the contextual features that influence parking occupancy.
- We also explored the relationship between area division and parking prediction. Moreover, we found that a shape-based clustering technique could boost the parking occupancy forecast accuracy when combined with regression models.
- The experiments showed that the internal clustering evaluation metrics are not suitable for spatio-temporal clustering if we aim to predict the trends of each cluster.

The rest of this paper is organised as follows: Section 2 summarises the literature review; Section 3 introduces the proposed framework, methodologies, and some terminologies in this paper; Section 4 presents the case study in the Melbourne CBD area, including descriptions of the multiple data sets, results of the experiments, and findings. Finally, Section 6 concludes the paper and discusses possible future work.

2. Literature review

Parking availability information can effectively aid drivers in searching for vacant parking slots so that driver frustration can be alleviated, and queues of waiting for parking, travelled distance, traffic congestion and air pollution are reduced [17]. The most prevalent way of finding parking positions by drivers is to search the nearest available parking slot at their destinations. However, it is easily to raise the competition for the most convenient alternative, which may cause long waiting time and increase the final searching time [18]. Estimating the occupancy of parking slots has become increasingly essential for guiding drivers to stop their cars at optimal positions. Recently, with the usage of more information systems and various data mining technologies, real-time parking availability information can be estimated in advance and provided to drivers, which can increase the overall success rate of parking vehicles in desire time [19]. In 2007, Caliskan et al. [20] predicted parking occupancy by analysing the flows of vehicles at parking slots and estimating the time of car arrivals with a continuous time-homogeneous Markov model. Caicedo et al. [18] presented a real-time availability forecast (RAF) algorithm to predict the number of expected market share in one parking facility at specific moments. This methodology contains three subroutines: parking request allocations, future departures estimations, and parking availability predictions. The results indicate that the approach can yield a small average error in parking availability of predictions, and the timing is an important factor in evaluating the efficiency of finding available parking spots. In 2014, Klappenecker et al. [21] also utilised a continuous-time Markov chain to predict the number of available parking spaces but simplified the computation of the transition probabilities for the navigation system.

Clustering methods have been frequently used in assisting the prediction of parking availability. Richter et al. stated that up to 30% of traffic is generated by vehicles that could not efficiently locate free parking spaces [22]. They presented an approach to learn the future parking availability from historical data and embedded these models within in-vehicle navigators. The results showed that the storage requirement of models can be decreased dramatically by clustering while ensuring good accuracy of the predictions. Vlahogianni et al. [23] segmented the sensor network of on-street parking into four different regions and applied an optimised Multilayer Perceptron (MLP) to predict parking occupancy in each region up to one hour in the future using the past five-minute data.

In recent years, various models of deep neural networks are becoming popular for parking availability prediction. Taking advantage of massive real-time parking data that obtained from parking sensor network in the city of Santander, Spain, Vlahogianni et al. [24] used survival and neural network models to predict the parking occupancy, along with the probability of finding free parking spaces. The experiment showed that simple, genetically optimised MLP could accurately predict region parking occupancy rates in the next 30 min using only one-minute data. With the consideration of time-of-day parking distribution and the user cost of parking (walking, cruising and parking fee), Lim et al. [25] presented an alternative model to improve the accuracy of the prediction of parking volume and provided more detailed information of the accessibility of parking facilities. In 2018, Shao et al. [9] employed the LSTM to predict parking availability in Melbourne City central area; the empirical results showed that such time-dependent neural network outperforms existing machine learning methods and other deep learning solutions. In contrast to relying on historical parking dataset, Rong et al. [26] predicted parking availability via the LSTM to model the temporal closeness and period influence in parking bays, with the consideration of several specific features, such as weather, holiday, human mobility and navigation data, and PoIs. Zhang et al. [27] mentioned that periodic parking patterns from historical records are an essential factor for parking behaviour prediction. They integrated weather information and parking changes data into a modified LSTM that could then learn periodic parking patterns from historical records, such as the same time in previous days, weeks and months.

Previous works have tended to estimate parking occupancy for all the targeted parking slots with historical parking data or environmental information, such as weather and human mobility data. In this paper, we mainly focused on using the contextual information to make the prediction of parking availability in certain areas (e.g. newly-built ones) where lack historical parking data or may not have datasets related to human activities. At the first stage, different combinations of clustering methods and regression models were assessed and compared in the prediction accuracy of parking occupancy. Then, based on the optimal clustering result, important features were identified through quantitative correlation analysis between various contextual features and parking occupancy, which could be utilised as inputs for a regression model to make the predictions without any historical data.

3. Methodology

3.1. Problem definition

In this paper, we studied two different problems: to predict parking occupancy using historical data and to predict parking occupancy using contextual features.

Parking occupancy using historical data is a classic time-series prediction problem, that is, predicting the most likely traffic measurements in the next H time steps given the previous M parking occupancy observation as: $\hat{o}_{t+1}, \dots, \hat{o}_{t+H} = \arg \max_{o_{t+1}, \dots, o_{t+H}} \log P(o_{t+1}, \dots, o_{t+H} | o_{t-M+1}, \dots, o_t)$, where $o_t \in \mathbb{R}^n$ is an observation vector of n parking spaces at time step t , each element of which records historical observation for a single parking node. Considering that each node can only contain one car, $\hat{o}_t$ can only be 0 or 1. In our proposed model, parking slots were grouped into k clusters based on the similarity of parking occupancy distributions. We aimed to predict the overall parking occupancy in the next H time steps, given the results of previous M parking occupancy observations as:

$$\hat{o}_{t+1}^i, \dots, \hat{o}_{t+H}^i = \arg \max_{o_{t+1}^i, \dots, o_{t+H}^i} \log P(o_{t+1}^i, \dots, o_{t+H}^i | o_{t-M+1}^i, \dots, o_t^i) \quad (1)$$

where i denotes the index of the cluster. Parking occupancy prediction using contextual features is different from the problem above. It is a typical supervised regression learning problem. We aimed to extract a set of features, $\mathbf{x}$ as the input. The feature space $\mathcal{X} \in \mathbb{R}^m$ is the space of all feasible contextual features. The label space $\mathcal{Y} \in [0, 1]$ is the domain of parking occupancy predictions. Let $\mathcal{D} : \mathcal{X} \times \mathcal{Y}$ be the probability distribution on the space of samples representing the contextual features of the prediction area. The goal of learning was to minimise the loss, as represented below:

$$\ell(h; \mathcal{D}) = \text{Exp}_{(x,y) \sim \mathcal{D}}[\ell(h; (x, y))], \quad (2)$$

where ℓ denotes the expected loss, Exp denotes the expectation, and the hypothesis $h : \mathcal{X} \rightarrow \mathcal{Y}$ is a mapping of input contextual features to predictions.

3.2. Experiment schema

Algorithm 1 k -Shape and LSTM

Require: Parking occupancy observation, cluster number k (e.g. $k = 29$), predictions horizon T
Ensure: Average Parking Occupancy $\bar{p}_i$
 Occupied probability of parking slots $o_i \leftarrow$ sampling parking occupancy observation
 With the value of k , apply k -shape clustering algorithm to parking slots to obtain cluster $C(|c| = 29)$
for t in T **do**
 for each cluster C_i , $0 < i \leq k$ **do**
 create and fit a time-series prediction neural network
end for
 Calculate Average Parking Occupancy $\bar{p}_i$ for each cluster C_i
end for

Algorithm 2 The feature-based model

Require: Parking availability forecast results, and contextual information.
Ensure: Parking occupancy of a new area P_{new} .
 Clustering parking slots using historical data.
 Get a set of features X
 Analyse Pearson correlation $\rho(p, x)$ between parking occupancy P and X
 choose $x_i \in X$ which $\rho(p, x) > T$ where T is the threshold of Pearson correlation
 Predict the parking occupancy of a new area using x_i

Algorithms 1 and 2 show our experiment schema consisting of two sessions: predicting the parking occupancy using historical data only and predicting the parking occupancy using contextual features.

In the first session, we applied different clustering algorithms combined with different regressors to the spatial information and temporal information, respectively. Then we obtained the parking occupancy of each sub-area. Although Shao et al. indicated that area division has a significant influence on parking occupancy forecast, the way to evaluate such division remains unclear [9]. Dividing the parking areas into sub-areas based on the parking situation of each parking lot is not a spatial segmentation problem. It needs to consider both spatial and temporal information. Therefore, it is a typical spatio-temporal clustering problem. Some researchers suggest that conventional clustering evaluation metrics cannot measure the performance of this kind of clustering methods because spatial and temporal information are located in different Euclidean spaces [28]. However, they did not propose a very clear example to validate the claim. Therefore, we compared different clustering methods and selected the best one as the area division approach. We also evaluated many existing regression models and selected the most suitable one for the chosen clustering algorithm.

In the second session, we compared the parking availability forecast results of each cluster and associated contextual information to infer and analysed the correlations. For each sub-area, we analyse the linear correlation between parking occupancy and all contextual features using the F-test. Then, we predicted the parking occupancy of a new area using all high-correlated contextual features rather than historical parking data. We also conducted an experiment to show that a traditional internal clustering evaluation metric does not work well in spatio-temporal clustering methods for prediction purposes.

3.3. Data preprocessing and analytical methods

In this section, we describe how we extracted features from contextual information and preprocessed the historical parking data. All data processing code and corresponding data will be published on GitHub.

Table 1
The details of dataset.

Total rows	21799521
Source domain	data.melbourne.vic.gov.au
Created	2017/11/28 01:03:42
Last updated	2019/4/9 14:20:02
Category	Transport
Attribution	City of Melbourne
License	Creative Commons Attribution 4.0 International
Owner	City of Melbourne Open Data Team

Table 2
The description of contextual features.

Features	Description
xP	x hours of free parking
Disable_Zone	You can only park in an area for people with disabilities
Loading_Zone	Only eligible vehicles can use Loading Zones when picking up or dropping off passengers or goods.
Permit_Zone	You can only park in a Permit Zone if you correctly display the permit specified on the sign.
No_Stop	No Stopping signs means you cannot stop for any reason or any length of time.
Nearest_Garage_Influence	Near the Garage
Nearby_Restaurants_Open_Distribution	Near the Restaurants which open distribution
Nearby_Unique_Office	Near the Unique Office
Nearby_Train	Near the Train
Nearby_Unique_Restaurants	Near the Unique Restaurants
Free_Tram_Zone_Ratio	The tram as a free transportation mode
Nearby_Bus	Near the Bus
Nearby_Tram	Near the Tram

3.3.1. Clustering parking slots using historical data

The dataset we used was collected from the Melbourne open data platform historical parking record [16]. This dataset is about on-street car parking sensor data in the city of Melbourne for the 2011 calendar year. The City of Melbourne has installed in-ground parking bay sensors in most CBD parking bays. These sensors record when a vehicle arrives and when it departs. Each record also includes the parking restriction for a bay and whether the vehicle has overstayed that restriction. Some details about the dataset is shown in Table 1. We split the November dataset into weekly, weekday and weekend datasets. To measure the similarity between the parking trends of different parking slots, we propose a new feature to represent the parking trend of each parking slot. First, we sampled the parking state of each parking slot every five minutes, where the state of a parking slot $v_i^t \in \{0, 1\}$ denotes the availability of the parking slot $v_i \in \mathbf{v}$ at time step t , where $\mathbf{v}$ represents the vector of parking occupancy of clusters. Second, we calculated a_i , the average of v_i , at the same time t in different days. a_i is considered as a feature of parking slot. Finally, we obtained n feature vectors $\mathbf{f}_{v_i} = [a_1, a_2, \dots, a_{288}]^T$ for one day, where n is the number of parking lots, and 288 is the number of samples ($60 \times 24 \div 5$). Subsequently, we applied the selected clustering algorithm to grouping parking slots with similar trends. Next, based on these clusters, the parking occupancy rate was calculated. Then, we applied the regression model to each cluster and used the three previous parking occupancy samples to predict the next 5-, 15-, or 30-min occupancy rate. The weekly forecast is based solely on historical data from a single week, which means that the data from the previous week will not be used to predict the current week's outcomes. For instance, when making a forecast for the second week, only the historical data from the second week will be considered, and the data from the first week will not be included in the analysis.

3.3.2. Extracting features from each cluster

In this paper, we propose 18 contextual features [27] Table 2 that may influence parking occupancy, and obtain them from the AURIN(Australian Urban Research Infrastructure Network) [29] and Google Map for each cluster.

The entire dataset contains 5280 records ($22 \text{ weekdays} \times 24 \text{ h} \times 10 \text{ clusters}$). We aimed to predict the parking occupancy in a new area by using only these features.

3.3.3. Extracted features from admin areas

This dataset consists of the same 18 features extracted from each admin area Table 2. This dataset is used to compare with the historical average prediction method to further prove the validity of these proposed features. Given that the historical average method uses last week's data to predict the next, we considered the parking records for the last week of October 2011. Therefore, this dataset contains 19,488 ($28 \text{ weekdays} \times 24 \text{ h} \times 29 \text{ areas}$) records, and the label is also parking occupancy rate.

4. Case study

4.1. Dataset description

Historical parking record datasets were collected from the Melbourne open data platform. We conducted all experiments on the on-street car parking sensors data in 2011.

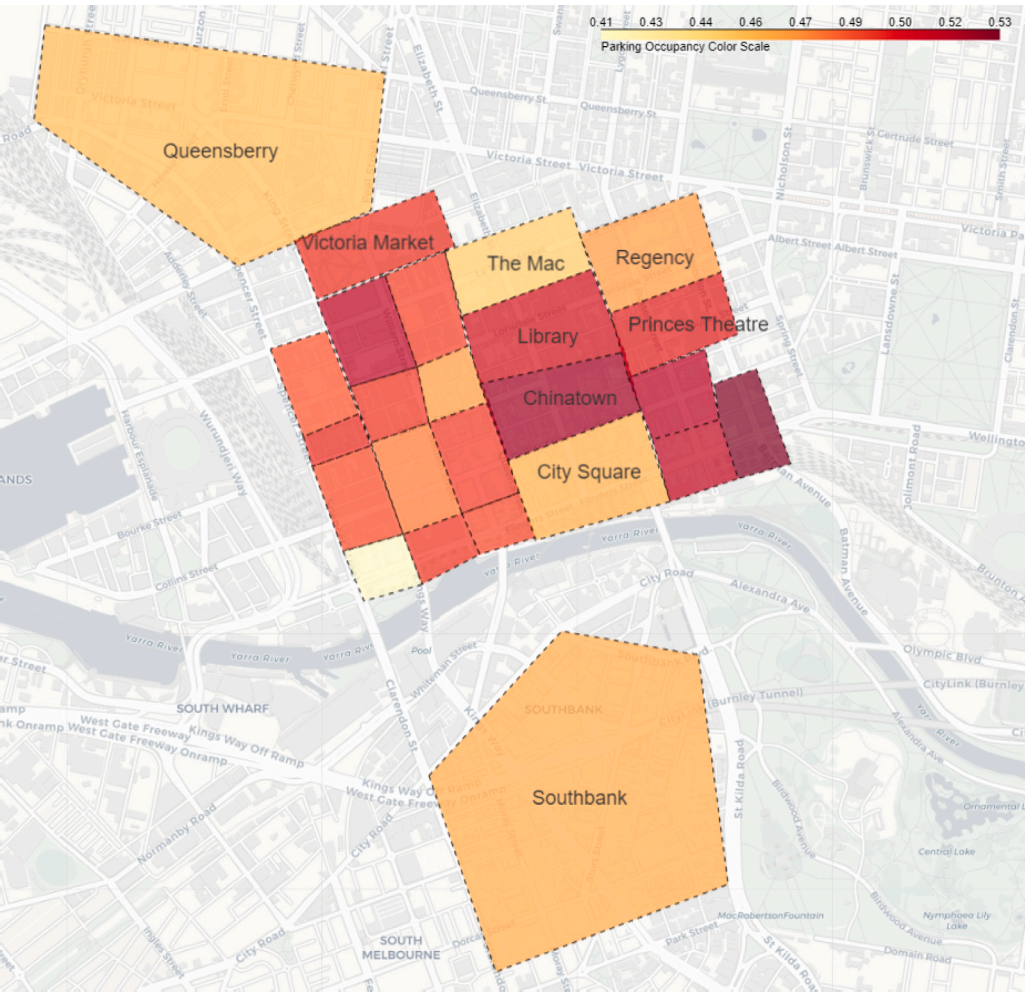


Fig. 1. Parking occupancy rate by areas in Melbourne CBD.

Table 3
Description of parking record.

Headers	Description
Administration area	The administration sub-region that a parking lot belongs to.
Arrive time	The start time of a parking event.
Departure time	The end time of a parking event.
Parking rules	The parking rule for that specific parking lot.
Longitude	Geographical information.
Latitude	Geographical information.

The parking system logs the arrival time and departure time of the car on each parking bay. The records also include the coordinates and parking rules of the bays. The period of the data used in this paper was from 24 July 2011 to 30 August 2011 and from 24 October 2011 to 30 November 2011, which has a total of 4156 parking lots. We evaluated the proposed two-step model and the new features for parking occupancy prediction with these data. All attributes of the parking record are listed in Table 3.

Fig. 1 illustrates the average parking occupancy rate of each sub-region in Melbourne CBD. The parking occupancy rate ranges from approximately 41% to 53%. Some sub-regions are similar in the geospatial aspect, but the difference in parking occupancy is large. Moreover, some sub-regions are far away on the map but have a similar average parking occupancy rate, which provided us with a chance to study the potential environmental factors that might influence parking events on the streets.

4.2. Experimental settings

4.2.1. Evaluation metric

In this section, we show the all evaluation metrics used in this paper. Two evaluation metrics were used: one is for a regression model, and the other is for a clustering analysis.

For the regression task, we applied three different metrics in this paper:

Mean Absolute Error (MAE) is the average of the difference between the true values and the predicted value.

$$MAE = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i| \quad (3)$$

where y_i is the true parking occupancy value, $\hat{y}_i$ is the predicted parking occupancy value and m is the number of parking lots.

Root Mean Square Error (RMSE) measures the standard deviation of the residuals.

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m y_i - \hat{y}_i} \quad (4)$$

where y_i is the true parking occupancy value, $\hat{y}_i$ is the predicted parking occupancy value, m is the number of parking lots.

Mean absolute percentage error (MAPE) which is similar to MAE, except that it computes the percentage error, which is more intuitive.

The second set of evaluation metrics was used for clustering results validation. Evaluation of clustering results is a challenging research problem in data mining [30]. The evaluation metric can be classified into two categories: internal and external evaluation. Internal evaluation indicates the quality of clustering results using a single value without any external data, such as labels or ground truth. External evaluation metric measures the quality of the clustering result using ground truth made by human experts. In our case, finding an expert to group parking slots based on the parking occupancy distributions was impossible. Therefore, we applied three internal clustering evaluation metrics (Davies–Bouldin index [31], Calinski and Harabaz (CH) [32] score, and Silhouette Coefficient [33]) as shown in [28], to assess the quality of the clustering methods.

To validate if the traditional clustering methods perform well on our problem, we also used the prediction results as the final evaluation metric to measure two different ways to group parking slots. First, we applied a shape-based clustering method to group parking slots. Second, we used the sub-regions designed by the City Council. If the classic internal clustering methods could measure the performance of each clustering method in our problem, both evaluation results should be the same. Otherwise, it suggests that the internal clustering evaluation metrics do not fit to problem.

4.2.2. Hyper-parameters

The prediction horizons were 5, 15, and 30 min. The batch size was 1. The epoch was 300. The training set to test set ratio was 7 to 3.

4.2.3. Clustering and regression algorithms

Our two-step model consists of two parts: clustering and regression. To determine the best selection for prediction purpose, we tested several regression models, including Linear Regression, Support Vector Regression Machine [34], Random Forest [35], Multi-layer Perceptron (MLP) [36] and Long short-term memory (LSTM) [37] on a one-month dataset.

LSTM is one of the most popular variants of Recurrent Neural Networks (RNNs), which usually process sequential data. It is capable of learning long-term dependencies, as it addresses issues, such as vanishing and exploding gradients, in the training stage of simple RNNs. As can be observed from Table 4, LSTM consistently achieves the best performance with most cases, which demonstrates the superiority of LSTM in accurately capturing time dependency. More specifically, the average prediction error (RMSE) of LSTM are 0.0436, 0.0913, and 0.1169 at 5, 15 and 30-min horizons, respectively.

To validate the usage of clustering techniques in spatio-temporal prediction, we compared a couple of representative traditional clustering approaches, including k -shape [38], K-Means [39], Fuzzy C-means [40], Gaussian Mixture Model [41] and Spectral clustering [42] with LSTM to predict the multi-step parking occupancy over a month. k -shape is a novel time-series clustering method that can preserve the shape information of time-series sequences. Similar to K-Means, an iterative assignment and refinement procedure is performed in this algorithm. However, unlike K-means, which treats the observations in time-series data as independent attributes, k -shape considers the time order and the shape of the time-series data during clustering. Specifically, it uses a normalised, domain-independent version of the cross-correlation measure as its distance measure, which is used to measure the similarity of time-lagged signals. In fact, we note that previous similar work [15] has used k-means for clustering without providing a clear justification for the choice, and they did not involve predicting future occupancy. Therefore, the benefits of the clustering method were not adequately demonstrated in their work. Nevertheless, we employ the structure of their algorithm and compare it with our proposed work, and find the result of [15] is similar to our result but our approach is still higher in general.

To demonstrate the superiority of our work compared to previous similar studies, we combined the method used in previous research with LSTM as a baseline.

Tables 5 and 6 show that k -shape outperforms other clustering algorithms in these datasets. More specifically, 5, 15 and 30 min, the combination of k -shape and LSTM achieves 0.0338, 0.0676 and 0.0869 in RMSE, respectively.

At the last stage, we selected the most suitable clustering and regression combination, k -shape and LSTM to predict the parking occupancy using historical data, where $k = 29$ as the number of admin areas. Fig. 2 illustrates that the two-step model outperforms the regression models without clustering significantly over different datasets.

Table 4

Evaluation (RMSE) on multi-horizon parking occupancy prediction without clustering (best performance displayed in bold).

Datasets	Prediction horizon	Regression models				
		LSTM	MLP	Random forest	Linear regression	SVR
Nov. week 1	5	0.0434	0.0439	0.0488	0.0448	0.1000
	15	0.0797	0.0870	0.0728	0.1120	0.0980
	30	0.1028	0.1234	0.1075	0.1565	0.0993
Nov. week 2	5	0.0441	0.0437	0.0468	0.0447	0.0999
	15	0.0939	0.0982	0.0941	0.1131	0.1103
	30	0.1075	0.1211	0.1072	0.1577	0.1086
Nov. week 3	5	0.0436	0.0437	0.0475	0.0449	0.0963
	15	0.0957	0.1004	0.0961	0.1118	0.1240
	30	0.1353	0.1134	0.1428	0.1571	0.1342
Nov. week 4	5	0.0435	0.0436	0.0470	0.0448	0.0962
	15	0.0959	0.1005	0.1032	0.1119	0.1262
	30	0.1352	0.1412	0.1584	0.1573	0.1348
Average	5	0.0436	0.0437	0.0475	0.0448	0.0981
	15	0.0913	0.0965	0.0915	0.1122	0.1147
	30	0.1169	0.1248	0.1289	0.1572	0.1192

Table 5

Evaluation (RMSE) on multi-horizon parking occupancy prediction in November with combinations of clustering methods and LSTM (best performance displayed in bold).

Datasets	Prediction horizon	Cluster model + LSTM					
		Raw	<i>k</i> -shape	K-means	Fuzzy-Cmeans	GMM	Spectral
Nov. week 1	5	0.0434	0.0358	0.0464	0.0359	0.0361	0.0431
	15	0.0797	0.0572	0.0690	0.0729	0.0682	0.0761
	30	0.1028	0.0748	0.0867	0.1012	0.0908	0.0937
Nov. week 2	5	0.0441	0.0398	0.0436	0.0433	0.0419	0.0479
	15	0.0939	0.0558	0.0692	0.0643	0.0765	0.1030
	30	0.1076	0.0867	0.0929	0.0903	0.0911	0.1295
Nov. week 3	5	0.0436	0.0478	0.0417	0.0497	0.0422	0.0440
	15	0.0957	0.0765	0.0774	0.0803	0.0812	0.0799
	30	0.1353	0.0962	0.0986	0.1111	0.1061	0.1025
Nov. week 4	5	0.0435	0.0407	0.0518	0.0386	0.0456	0.0445
	15	0.0959	0.0632	0.0804	0.0816	0.0827	0.0826
	30	0.1352	0.0834	0.0987	0.1113	0.1075	0.1035
Average	5	0.0436	0.0435	0.0484	0.0444	0.0440	0.0449
	15	0.0913	0.0632	0.0740	0.0748	0.0772	0.0854
	30	0.1202	0.0853	0.0942	0.1380	0.0989	0.1073

4.2.4. Internal cluster analysis

To examine the effectiveness of the *k*-shape cluster, we use the *Davies Bouldin score*, Calinski Harabaz score and Silhouette score metrics to measure the clustering results using the *k*-shape clustering and the administration area. The City Council designed the administration area according to the demographic information and policy, and the *k*-shape-based clustering results were divided based on the shapes of distribution of each parking slot occupancy.

For each clustering result, we applied the LSTM to predict the parking occupancy of each cluster and measure the average regression error rate in Table 7. The results show that administration areas are a better clustering result over three different internal clustering evaluation metrics. However, the prediction error rates show the opposite results that *k*-shape clustering perform significantly better than the administration area, which concludes that the conventional internal clustering is not good at validating the quality of clustering for prediction purpose in spatio-temporal data.

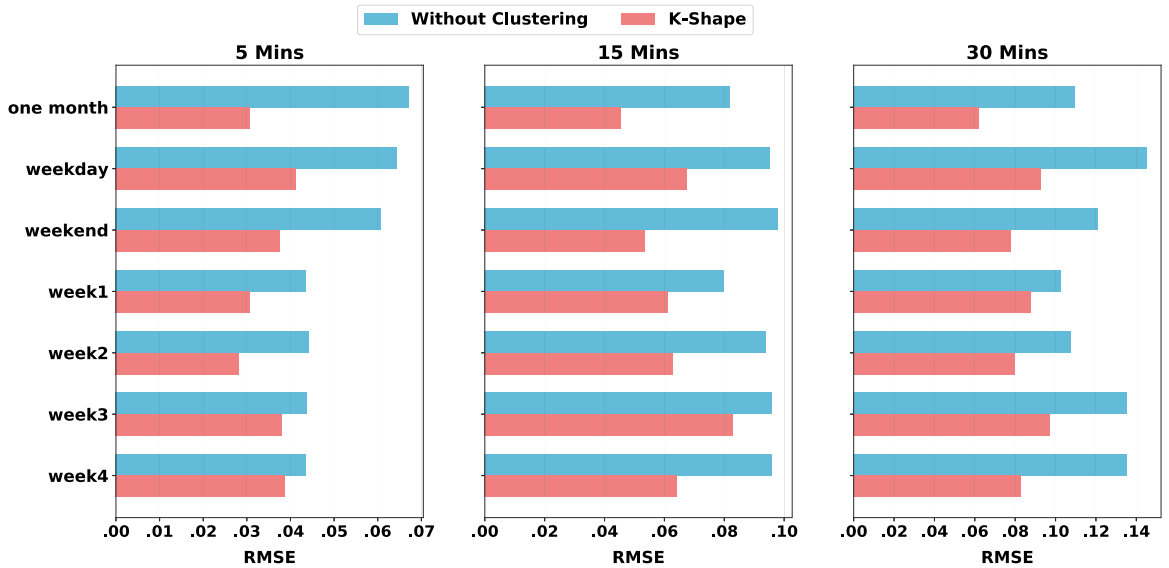
4.3. Feature analysis

In this section, we demonstrate the detailed procedures in analysing and verifying the validity of each contextual feature. Given that some extracted features are only valid in the daytime, most of the analyses are based on the comparison between the values of features and the average daytime parking occupancy rate (7 am to 7 pm). For better observation of the visualisation results, we used Min-Max normalisation to convert all features to a [0, 1] scale. In addition, cluster 5 only contains two parking lots which are located in front of the main entrance of one auto repair shop. Given that almost no one would park their car in front of the entrance of stores, we regarded this cluster as an outlier and ignored it in the following analysis.

Table 6

Evaluation (RMSE) on multi-horizon parking occupancy prediction in August with combinations of clustering methods and LSTM (best performance displayed in bold).

Datasets	Prediction horizon	Cluster model + LSTM					
		Raw	<i>k</i> -shape	K-means	Fuzzy-Cmeans	GMM	Spectral
Nov. week 1	5	0.0629	0.0377	0.0494	0.0448	0.0503	0.0434
	15	0.0823	0.0489	0.0662	0.0603	0.0658	0.0555
	30	0.1015	0.0619	0.0813	0.0744	0.0812	0.0702
Nov. week 2	5	0.0588	0.0467	0.0492	0.0470	0.0505	0.0521
	15	0.0761	0.0558	0.0692	0.0643	0.0765	0.1030
	30	0.0908	0.0689	0.0750	0.0726	0.0816	0.0818
Nov. week 3	5	0.0653	0.0506	0.0536	0.0564	0.0522	0.0501
	15	0.0841	0.0639	0.0698	0.0731	0.0673	0.0657
	30	0.1059	0.0756	0.0885	0.0871	0.0847	0.0824
Nov. week 4	5	0.0616	0.0408	0.0411	0.0493	0.0448	0.041
	15	0.0829	0.0522	0.0521	0.0663	0.0569	0.0541
	30	0.1024	0.0638	0.0662	0.0819	0.0694	0.0661
Average	5	0.0621	0.0440	0.0483	0.0494	0.0495	0.0467
	15	0.0814	0.0557	0.0624	0.0652	0.0639	0.0602
	30	0.1002	0.0676	0.0778	0.0790	0.0792	0.0751

**Fig. 2.** Two-step model validation on different datasets.**Table 7**

Internal cluster analysis.

Evaluation metrics	Clustering area	Admin area
davies_bouldin_score	13.5566	4.2717
calinski_harabaz_score	39.0585	408.2195
silhouette_score	-0.5831	-0.4696
ine RMSE (5 min)	0.0338	0.0672
RMSE (15 min)	0.0676	0.1166
RMSE (30 min)	0.0869	0.1563

We calculated three metrics between each feature and the average occupancy rate: Pearson Correlation Coefficient (PCC), F value and *p*-value. The F value and *p*-value were computed via a Univariate Linear Regression Tests and its null hypothesis is that the feature and occupancy rate have no dependency. These three metrics were regarded as significant guidance for the feature selection in our experiments, and the values are displayed in Table 8. The experimental results show that all features are statistically significant correlated with the parking occupancy rate, as the *p*-value is less than 0.005.

Table 8
Features correlations (ordered by PCC values).

Features	Pearson correlation coefficient	F value	p-value
Disable_Zone	0.51	1940.91	0
2P	0.41	1086.93	6.44E-217
3P	0.41	1068.95	1.13E-213
1P	0.40	1051.97	1.33E-210
0.25P	0.33	667.93	8.80E-139
Loading_Zone	0.33	664.17	4.69E-138
0.5P	0.32	608.61	2.83E-127
4P	0.29	507.84	1.90E-107
Nearest_Garage_Influence	0.17	173.26	5.86E-39
Permit_Zone	0.16	142.04	2.46E-32
No_Stop	0.14	118.75	2.31E-27
Nearby_Restaurants_Open_Distribution	0.13	93.88	5.09E-22
Nearby_Unique_Office	0.11	70.95	4.64E-17
Nearby_Train	0.11	68.40	1.67E-16
Nearby_Unique_Restaurants	0.08	39.31	3.88E-10
Free_Tram_Zone_Ratio	-0.05	16.40	5.18E-05
Nearby_Bus	-0.37	857.94	7.00E-175
Nearby_Tram	-0.40	1030.53	1.04E-206

4.3.1. Influence of commercial off-street parking lots

Most cities usually have two types of parking zones. One is a public parking zone, such as on-street parking slots, the other is a commercial parking zone, such as underground shopping mall parking zones. In this study, we aimed to analyse the correlation between these two types of zones. We find that the price of two different parking zones plays a crucial role in the parking occupancy of on-street parking slots. This feature is defined as the ratio of the hourly price difference between on-street and off-street parking lots to the distance between them. Mathematically, it is represented as follows:

$$\text{Nearest_Garage_Influence} = \frac{\sum_{p_i^* \in \Gamma(p_i)} |\$(p_i^*) - \$(p_i)|}{d^2}, \quad (5)$$

where d denotes the distance between the on-street parking slot p_i and the off-street parking slot p_i^* within the neighbourhood $\Gamma(p_i)$. Here, we define the neighbourhood $\Gamma(p_i)$ as the set of off-street parking slot p_i^* , $d(p_i, p_i^*) < 200$ m.

A larger value indicates that the gap between the price of the nearby off-street parking slots and the on-street slots is wider. Hence, we assumed that a higher ratio would lead to a higher on-street parking occupancy in each area. We examined the relationship between the daytime occupancy rate and the influence of off-street parking lots in each cluster. The PCC of this feature is 0.1783, the F value is 173.2621, and p -value is 5.8624E-39. Furthermore, according to the results shown in Fig. 3, a positive relationship exists between the occupancy rate and the off-street parking lots influence in clusters 0, 1, 3, 4, 6, 7, 8, and 9. Although cluster 2 does not obey this regular pattern, the reason may be that the number of restaurants around cluster 2 (430 restaurants) is much larger than that of clusters 1 (82 restaurants) and 3 (139 restaurants). Furthermore, these three clusters contain a tiny amount of sensors, where cluster 1 has 3 sensors (0.06% of the amount of the total sensors), cluster 2 has 34 sensors (0.7%), and cluster 3 has 6 sensors (0.13%). As a result, we consider this feature as valid for distinguishing the occupancy rate.

4.3.2. Free tram zone ratio

Free tram zone is a unique feature of Melbourne. The Melbourne City Council has still preserved the tram as a transportation mode, and all passengers do not need to pay in the city areas, which are called “free tram zone”.

This feature is defined as the ratio of the number of parking lots belonging to the free tram area within a cluster to the number of total parking lots within a cluster. Mathematically, it is represented as follows:

$$\text{Ratio} = \frac{|\mathbb{F} \cap \mathbb{C}_i|}{|\mathbb{C}_i|}, \quad (6)$$

where $\mathbb{F}$ is the set of parking slots in the free tram zone, $\mathbb{C}_i$ denotes the parking slots in the cluster i , and $|\cdot|$ denotes the absolute number of the elements in the set.

Empirically, people tend to take a tram in the free tram zone instead of driving a car. Thus, we hypothesize that the higher free tram zone ratio of a cluster, the lower the occupancy rate. From the experiment, the PCC, F value and p -value of this feature is -0.0557, 16.4068 and 0.0005, respectively. The result in Fig. 4 does not show a clear relationship between this feature and the occupancy rate. Therefore, we suggest this feature has almost no effect on the occupancy rate.

4.3.3. Nearby unique office/restaurant amount

These two features calculate the number of unique offices and restaurants within a 200-m radius for every cluster separately. The PCC, F-value and p -value of the office number to the occupancy rate are 0.1152, 70.9598 and 4.6487E-17, and those of the restaurant number are 0.0860, 39.3175 and 3.8883E-10, respectively. In accordance with the results in Fig. 5, a positive relationship exists between these two features and the occupancy rate in clusters 0, 1, 2, 3, 7, 8, and 9. Cluster 4 is an exception because over

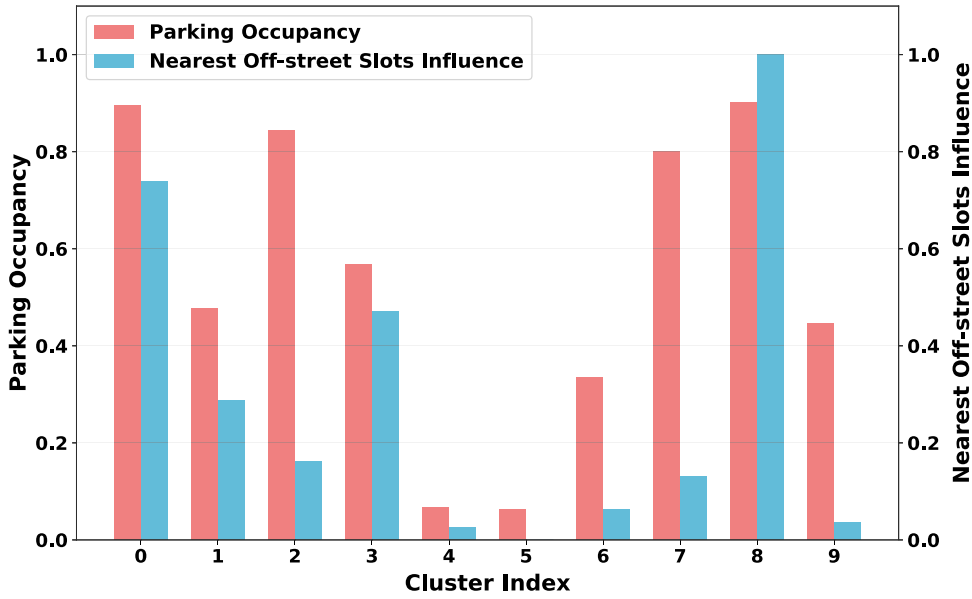


Fig. 3. Commercial off-street parking lots influence.

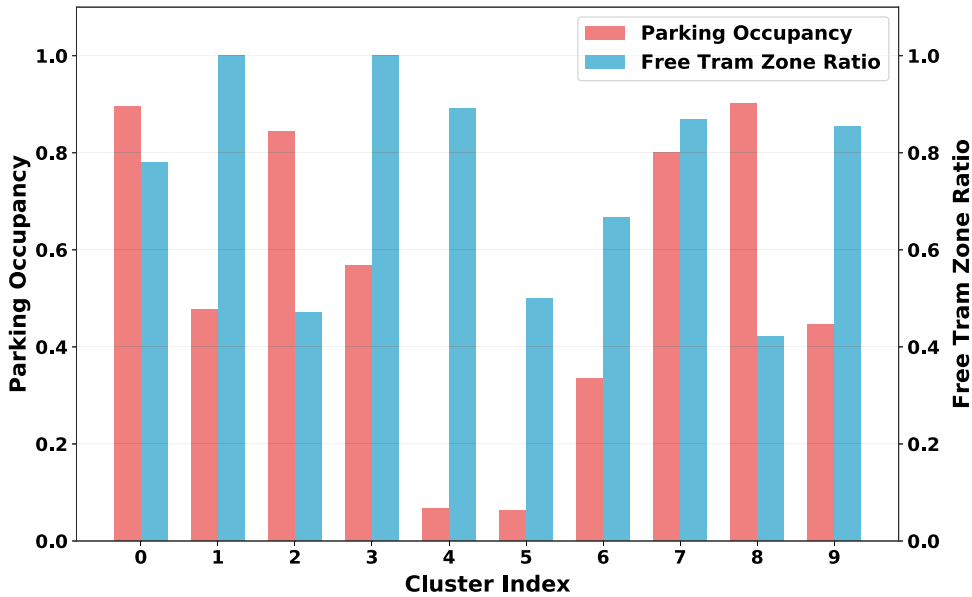


Fig. 4. Free tram zone ratio in different clusters.

50% of parking lots in cluster 4 are permit zones and disable zones (Disable zones are only available for disabled people.) between 7 am to 7 pm every weekday, which is illustrated in Fig. 8. In the light of Fig. 7, public transportation in cluster 4 is very developed compared with other clusters. These two reasons may largely reduce the parking occupancy rate as a result. Nevertheless, we still suggest that these two features are effective in occupancy rate prediction.

4.3.4. Nearby opening restaurants distribution

We investigated the number of opening restaurants within a 200-m radius of each cluster on an hourly basis and compared the trend of the amount of opening restaurants with the average occupancy rate trend in 24 h. The PCC, F value and p -value of this feature are 0.1322, 93.8856 and $5.0948E-22$, respectively. As shown in Fig. 6, the occupancy rate trends of clusters 0, 7, 8 and 9 are similar to the number of opening restaurants trend. Clusters 1 and 3 have stable occupancy rates at a lower level due to an over 85% proportion of permit zones and disable zones in these two clusters. Cluster 2 has a stable occupancy rate at a high level, given

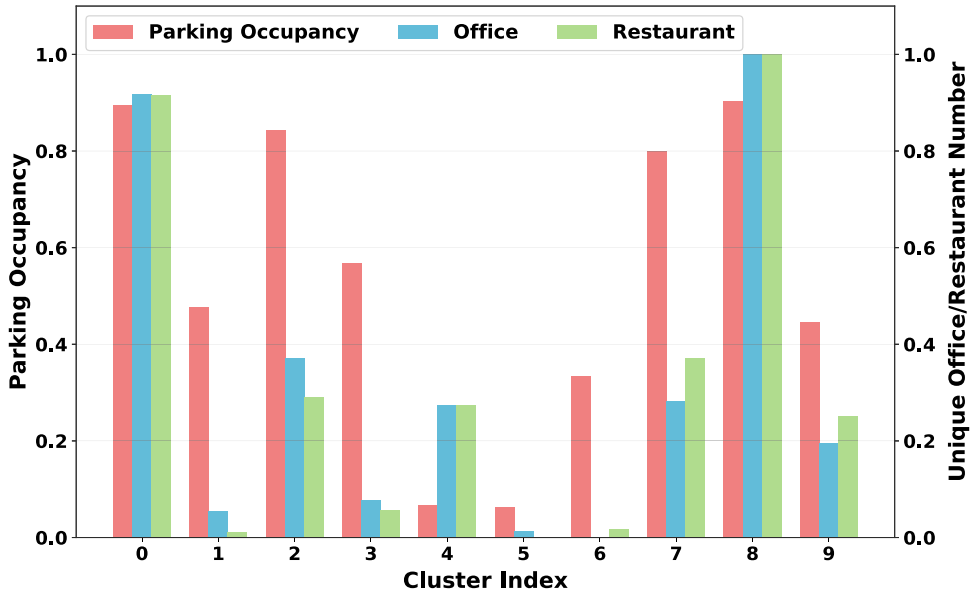


Fig. 5. Nearby unique office/restaurant number.

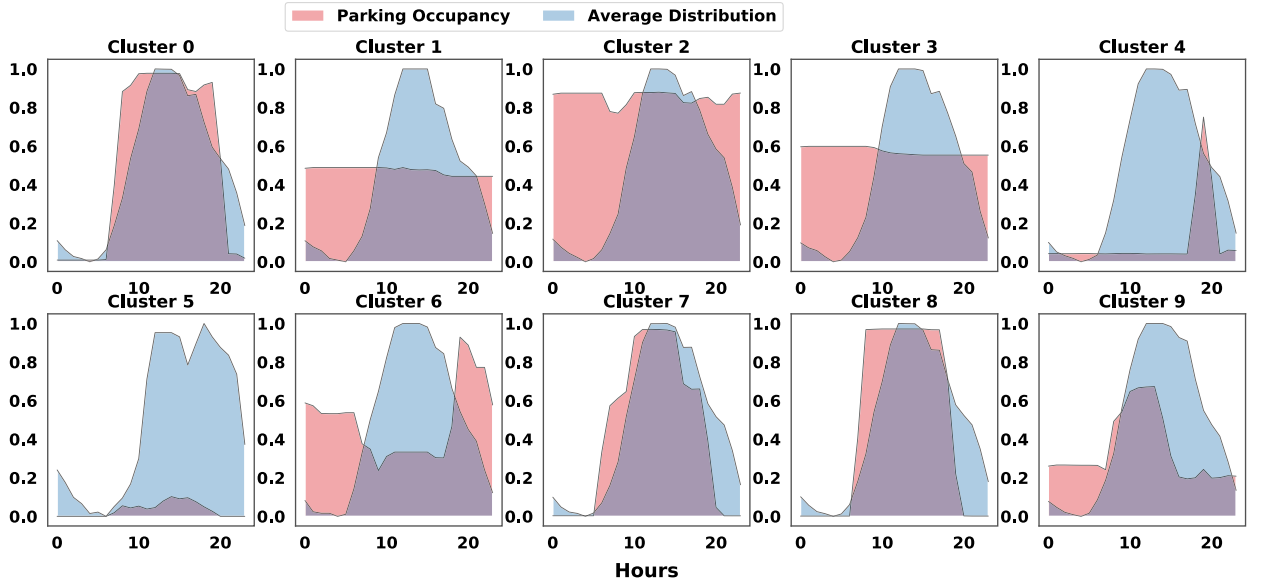


Fig. 6. Nearby restaurant opening hour distribution.

that most of the parking lots in this cluster are located at the edge of the free tram zone and are surrounded by many bars. Clusters 4 and 6 have low occupancy rates from 7 am to 7 pm because of the restriction of the permit zone and disable zone. After 7 pm, their occupancy rate trends are similar to the number of opening restaurants, as the rules are invalid after this time. One difference between these two clusters is that over 60% of the parking lots in cluster 6 are close to the grocery stores; hence, the occupancy rate in midnight is still around 0.5 because of the parking of trucks. Therefore, we could assume that the nearby opening restaurant distribution has a big impact on the parking occupancy rate.

4.3.5. Nearby public transportation distribution

We considered the influence of trams, trains and buses and calculated the average number of tram/train/bus stations within a 200-m radius of each cluster. The PCCs, F values and p-values of the number of tram/train/bus stations to the occupancy rate were $-0.4042/1030.5306/1.0468E-206$, $0.1131/68.4028/1.6710E-16$ and $-0.3739/857.9415/7.0077E-175$, respectively. Fig. 7 shows that more developed public transportation is likely to lead to a lower occupancy rate. This is very local because relatively

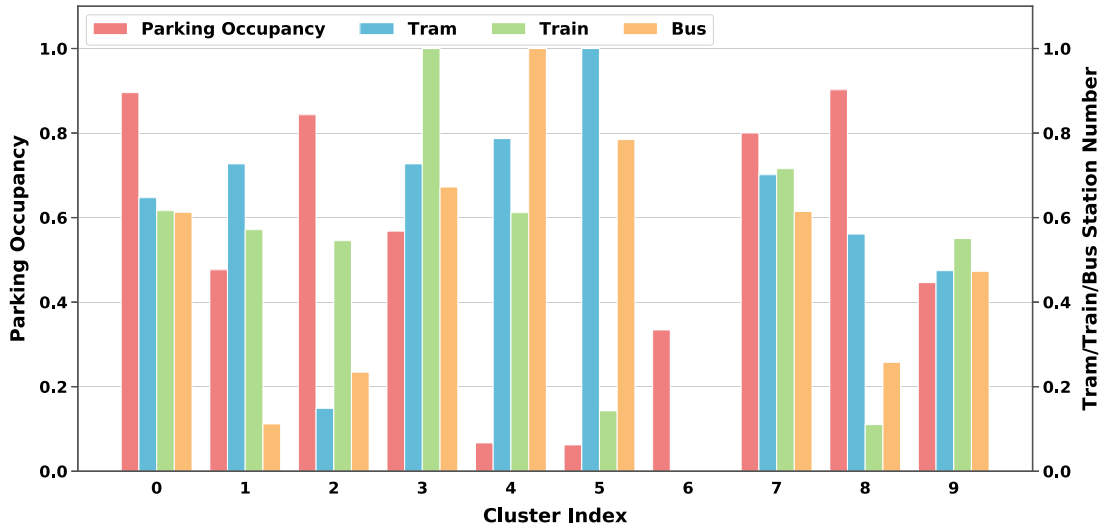


Fig. 7. Nearby public transportation distribution.

few parking spaces are located around tram/train/bus stops, and people's willingness to drive is not strong when going to an area with developed public transportation. Therefore, nearby public transportation is an important feature in occupancy rate prediction.

4.3.6. Parking rule distribution

Melbourne has ten different kinds of parking signs: 0.25P, 0.5P, 1P, 2P, 3P, 4P, Disable, Permit Zone, No Stop Zone, and Loading Zone. Given that most parking signs are valid from 7 am to 7 pm, we first calculated the number of different parking signs on an hourly basis for each cluster and compared the trend of the number of valid parking signs with the average occupancy rate trend in 24 h to obtain the PCCs/F values/p-values. Then, based on every parking rule's PCCs/F values/p-values, we selected the top six parking rules with larger PCCs and calculated the proportion of different parking rules in each cluster to determine the relationship between the rules and the occupancy rate. By comparing the average occupancy rate trend in Fig. 6 and the proportion of different parking rules in Fig. 8, we could observe that clusters 0, 2, 7 and 8 have very high occupancy rates during 7 am to 7 pm because of a large proportion of '1P', '2P', '3P' rules in these clusters. Clusters 1 and 3 have stable occupancy rates at a lower level due to over 85% proportion of permit zone and disable zone in these two clusters. Clusters 4 and 6 have a low occupancy rate from 7 am to 7 pm because of an approximate 50% proportion of the permit zone and disable zone within the clusters. The occupancy rate increases dramatically after the parking rules become invalid. As for cluster 9, most of the parking lots in this cluster are located at the corner or the end of the streets, which are secluded places. Thus, the highest occupancy rate of this cluster is less than 0.7, even though it contains over 90% proportion of the '1P' and '2P' rules.

In summary, according to our experimental results, parking rule distribution is the most significant feature that influences parking occupancy rates.

4.3.7. Comparison between feature prediction and nearby prediction

Based on the above feature analysis, we calculated the values of all the features, except the 'Free Tram Zone Ratio' for every administration area, and created the dataset 'Features of Admin Areas Dataset', which is mentioned in Section 3.3.3. Then, we used the LightGBM regression model [43] to predict the occupancy rate based on features and compare the results with the prediction results made by Historical Average. In previous works, there have been efforts to predict parking lot usage patterns based on context characteristics [15]. However, our work goes beyond that and focuses on predicting the future occupancy rate of the parking lot based on the same context characteristics. This has more practical significance as the classification of usage patterns is limited by the existing data, whereas the parking occupancy rate is not constrained by these limitations. In other words, the data available for usage patterns may not provide a complete or accurate representation, while the parking occupancy rate is not subject to the same limitations and may provide a more reliable indicator of actual usage. The results in Table 9 illustrate that the feature-based model outperforms the Historical Average method in all three metrics, which further validates the utility of our proposed features.

4.3.8. Comparison of the influence of features with high linear correlations

In this section, we undertake a systematic examination of potential feature collinearity and its influence on our model. We consider features demonstrating a Pearson Correlation Coefficient exceeding 0.1 or falling below -0.1, indicative of strong linear relationships. Notably, several features in our dataset are mutually exclusive. For instance, a location classified as a parking lot cannot simultaneously fall under the categories of a Disabled Zone, 1P, or 2P. Compound features, such as certain distributions, are also excluded from our analysis.

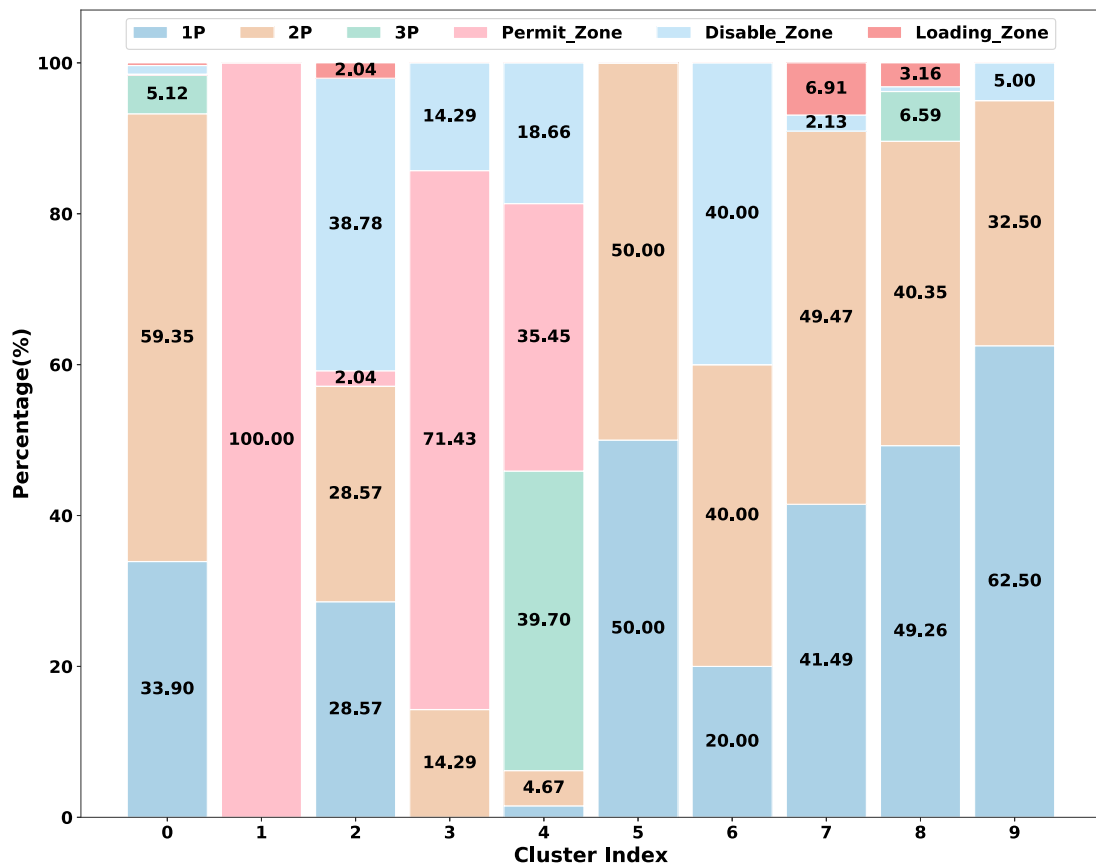


Fig. 8. Parking rules proportion.

Table 9

Comparison between two occupancy rate prediction methods using the admin area.

Evaluation metrics	RMSE		MAE		MAPE (%)	
	Features	Historical average	Features	Historical average	Features	Historical average
Nov. week1	0.0482	0.0960	0.0209	0.0857	2.74	10.98
Nov. week2	0.0501	0.1085	0.0218	0.0913	2.95	11.67
Nov. week3	0.0561	0.1321	0.0256	0.1131	3.42	17.52
Nov. week4	0.0568	0.1042	0.0261	0.0926	3.48	12.99
Average	0.0528	0.1102	0.0236	0.0957	3.14	13.29

Table 10

Evaluation (RMSE) on multi-horizon parking occupancy prediction in Nov without similar features.

Prediction horizon	Without NGI	Without NUO	Without NB	Without NT
5	0.0458	0.0452	0.0449	0.0457
15	0.0687	0.0667	0.0651	0.0671
30	0.0975	0.0943	0.0876	0.0901

With these selection criteria, the following features are singled out: Nearest Garage Influence, Nearby Unique Office, Nearby Train, Nearby Bus, and Nearby Tram. We subsequently computed the pairwise collinearity between these selected features.

To examine the impact of high-correlation features on prediction accuracy, we conducted further experiments excluding one of these features at a time. The Root Mean Square Error (RMSE) of multi-horizon parking occupancy prediction for November, excluding similar features, was calculated and is shown in Table 10:

The results suggest that the removal of one feature while retaining the others with high correlations does not dramatically influence the predictive outcome. Moreover, the decline in performance for more important features suggests a more pronounced impact when these are omitted. This understanding provides valuable insights into individual features' contributions to the model's overall performance.

Table 11
The influence of using historical data or not.

Different time period	Only use contextual features	Using both historical data and contextual features
Nov. week1	0.0675	0.0495
Nov. week2	0.0609	0.0578
Nov. week3	0.0675	0.0634
Nov. week4	0.0640	0.0523

4.3.9. Comparison of the influence of using historical data or not

It is infeasible to obtain historical data because the period we are predicting is before the parking lot is built. We proposed a method to predict parking occupancy using only contextual features without historical data. In Table 11, the result shows that it performs worse than using both historical data and contextual features. The processing of historical data is similar to that in Section 3.3.1, where the prediction for the second week be made without using any information from the first week, and the settings for other weeks are the same. The only use contextual features RMSE of week1, week2, week3 and week 4 are 0.0675, 0.0609, 0.0675, 0.064, less than 0.0495, 0.0578, 0.0634, 0.0523.

5. Discussion

Our study aimed to predict parking occupancy using historical data records in other areas and contextual information within the targeted area that lacks historical data. To achieve these goals, we proposed a two-step framework to learn the essential contextual features from the areas where parking records already exist and then transfer these features to the other new areas without historical data records. The results indicate that the k-shape and LSTM model and the feature-based model outperform the other models over different datasets.

Our study has some limitations. Firstly, this study concentrated on the common situation and predicted the parking records with historical data; hence, our model will not work when dealing with particular cases, for example, an event in the vicinity causing a sudden surge in visitors needing parking spots. Besides, our study lacked of other cities' parking records to validate our results further. Although most of the contextual features in our study can be applied to other cities, they have not been further verified by experiments.

Future studies, informed by the study we reported here, will extend the analysis to other cities to provide a more dynamic prediction for parking occupancy and extract more universal features. We will extend the analyses for prediction when occupancy rate predictions based on historical data go wrong (as in the example mentioned above) and provide a more dynamic, recent data-based parking occupancy prediction.

6. Conclusion

In this paper, we explore the different combinations of clustering methods and regression models in parking availability prediction with historical data. We also proposed and performed an in-depth analysis of relevant contextual features and estimated the potential parking occupancy in a new area using only highly correlated contextual features. We conducted extensive experiments with a large real-world dataset. The empirical results show that different clustering approaches have a significant influence on occupancy prediction accuracy. In the future, we will integrate the clustering methods with deep learning techniques and develop an end-to-end model to estimate the parking occupancy in a new area.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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References

- [1] J.N. Van Ommeren, D. Wentink, P. Rietveld, Empirical evidence on cruising for parking, *Transp. Res. A* 46 (1) (2012) 123–130.
- [2] D. Shoup, *The High Cost of Free Parking: Updated Edition*, Routledge, 2017.
- [3] D.C. Shoup, Cruising for parking, *Transp. Policy* 13 (6) (2006) 479–486.
- [4] E. Inci, A review of the economics of parking, *Econ. Transp.* 4 (1–2) (2015) 50–63.
- [5] X. Liang, B.O. Pérez, S.S. Dey, H. Haney, J.Y. Kim, Big-data analytics drives parking policy: Evaluating adherence to meter time limits in washington, DC, *Transp. Res. Rec.* 2594 (1) (2016) 107–117.
- [6] F.V. Monteiro, P. Ioannou, On-street parking prediction using real-time data, in: 2018 21st International Conference on Intelligent Transportation Systems, ITSC, IEEE, 2018, pp. 2478–2483.
- [7] A. Millard-Ball, R.R. Weinberger, R.C. Hampshire, Is the curb 80% full or 20% empty? Assessing the impacts of san francisco's parking pricing experiment, *Transp. Res. A* 63 (2014) 76–92.
- [8] W. Shao, F.D. Salim, T. Gu, N.-T. Dinh, J. Chan, Traveling officer problem: Managing car parking violations efficiently using sensor data, *IEEE Internet Things J.* 5 (2) (2017) 802–810.
- [9] W. Shao, Y. Zhang, B. Guo, K. Qin, J. Chan, F.D. Salim, Parking availability prediction with long short term memory model, in: *International Conference on Green, Pervasive, and Cloud Computing*, Springer, 2018, pp. 124–137.
- [10] T. Tiedemann, T. Vögele, M.M. Krell, J.H. Metzen, F. Kirchner, Concept of a data thread based parking space occupancy prediction in a berlin pilot region, in: *Workshops At the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015.
- [11] A. Tamrazian, Z. Qian, R. Rajagopal, Where is my parking spot? Online and offline prediction of time-varying parking occupancy, *Transp. Res. Rec.* 2489 (1) (2015) 77–85.
- [12] J. Xiao, Y. Lou, J. Frisby, How likely am I to find parking?—a practical model-based framework for predicting parking availability, *Transp. Res. B* 112 (2018) 19–39.
- [13] Y. Zheng, S. Rajasegarar, C. Leckie, Parking availability prediction for sensor-enabled car parks in smart cities, in: *IEEE Tenth International Conference on Intelligent Sensors, Sensor Networks and Information Processing*, ISSNIP, IEEE, 2015, pp. 1–6.
- [14] S. Yang, W. Ma, X. Pi, S. Qian, A deep learning approach to real-time parking occupancy prediction in spatio-temporal networks incorporating multiple spatio-temporal data sources, 2019, arXiv preprint arXiv:1901.06758.
- [15] K. Jayarajah, A. Tan, A. Misra, Understanding the interdependency of land use and mobility for urban planning, in: *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers*, 2018, pp. 1079–1087.
- [16] *Melbdata, City of Melbourne's Open Data Platform*, URL <https://data.melbourne.vic.gov.au>.
- [17] J. Cao, M. Menendez, R. Waraich, Impacts of the urban parking system on cruising traffic and policy development: the case of zurich downtown area, Switzerland, *Transportation* 46 (3) (2019) 883–908.
- [18] F. Caicedo, C. Blazquez, P. Miranda, Prediction of parking space availability in real time, *Expert Syst. Appl.* 39 (8) (2012) 7281–7290.
- [19] F. Caicedo, F. Robuste, A. Lopez-Pita, Parking management and modeling of car park patron behavior in underground facilities, *Transp. Res. Record* 1956 (1) (2006) 60–67.
- [20] M. Caliskan, A. Barthels, B. Scheuermann, M. Mauve, Predicting parking lot occupancy in vehicular ad hoc networks, in: *2007 IEEE 65th Vehicular Technology Conference-VTC2007-Spring*, IEEE, 2007, pp. 277–281.
- [21] A. Klappenecker, H. Lee, J.L. Welch, Finding available parking spaces made easy, *Ad Hoc Netw.* 12 (2014) 243–249.
- [22] F. Richter, S. Di Martino, D.C. Mattfeld, Temporal and spatial clustering for a parking prediction service, in: *IEEE 26th International Conference on Tools with Artificial Intelligence*, IEEE, 2014, pp. 278–282.
- [23] E.I. Vlahogianni, K. Kepaptsoglou, V. Tsetos, M.G. Karlaftis, Exploiting new sensor technologies for real-time parking prediction in urban areas, in: *Transportation Research Board 93rd Annual Meeting Compendium of Papers*, 2014, pp. 14–1673.
- [24] E.I. Vlahogianni, K. Kepaptsoglou, V. Tsetos, M. Karlaftis, A real-time parking prediction system for smart cities, *J. Intell. Transp. Syst.* 20 (2) (2016) 192–204.
- [25] H. Lim, G.T. Williams, D. Abdelqader, J. Amagliani, Z. Ling, D.W. Priester, C.R. Cherry, Alternative approach for forecasting parking volumes, *Transp. Res. Proc.* 25 (2017) 4171–4184.
- [26] Y. Rong, Z. Xu, R. Yan, X. Ma, Du-parking: Spatio-temporal big data tells you realtime parking availability, in: *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018, pp. 646–654.
- [27] F. Zhang, N. Feng, Y. Liu, C. Yang, J. Zhai, S. Zhang, B. He, J. Lin, X. Du, Pwllstm: Periodic LSTM with weather-aware gating mechanism for parking behavior prediction, in: *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, pp. 4424–4430.
- [28] W. Shao, F.D. Salim, A. Song, A. Bouguettaya, Clustering big spatiotemporal-interval data, *IEEE Trans. Big Data* 2 (3) (2016) 190–203.
- [29] U. of Melbourne, *Australian Urban Research Infrastructure Network*, URL <https://aurin.org.au/>.
- [30] D. Pfizner, R. Leibbrandt, D. Powers, Characterization and evaluation of similarity measures for pairs of clusterings, *Knowl. Inf. Syst.* 19 (3) (2009) 361.
- [31] D.L. Davies, D.W. Bouldin, A cluster separation measure, *IEEE Trans. Pattern Anal. Mach. Intell.* (2) (1979) 224–227.
- [32] T. Caliński, J. Harabasz, A dendrite method for cluster analysis, *Commun. Stat.-Theor. Methods* 3 (1) (1974) 1–27.
- [33] P.J. Rousseeuw, Silhouettes: a graphical aid to the interpretation and validation of cluster analysis, *J. Comput. Appl. Math.* 20 (1987) 53–65.
- [34] H. Drucker, C.J. Burges, L. Kaufman, A.J. Smola, V. Vapnik, Support vector regression machines, in: *Advances in Neural Information Processing Systems*, 1997, pp. 155–161.
- [35] A. Liaw, M. Wiener, et al., Classification and regression by randomforest, *R News* 2 (3) (2002) 18–22.
- [36] I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press, 2016.
- [37] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Comput.* 9 (8) (1997) 1735–1780.
- [38] J. Paparrizos, L. Gravano, K-shape: Efficient and accurate clustering of time series, in: *Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data*, ACM, 2015, pp. 1855–1870.
- [39] J.A. Hartigan, M.A. Wong, Algorithm AS 136: A k-means clustering algorithm, *J. R. Stat. Soc. Ser. C (Appl. Stat.)* 28 (1) (1979) 100–108.
- [40] J.C. Bezdek, R. Ehrlich, W. Full, FCM: The fuzzy c-means clustering algorithm, *Comput. Geosci.* 10 (2–3) (1984) 191–203.
- [41] G.J. McLachlan, K.E. Basford, *Mixture Models: Inference and Applications To Clustering*, vol. 84, M. Dekker New York, 1988.
- [42] A.Y. Ng, M.I. Jordan, Y. Weiss, On spectral clustering: Analysis and an algorithm, in: *Advances in Neural Information Processing Systems*, 2002, pp. 849–856.
- [43] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, Lightgbm: A highly efficient gradient boosting decision tree, in: *Advances in Neural Information Processing Systems*, 2017, pp. 3146–3154.